TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Technology that Enables Onboard Physical AI within the Computational Constraints of Humanoid Robots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Machinery and materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Robotics/Automation Machinery	Robotics/Automation Machinery Software
Concept of Proposed Technology	The proposed technology aims to enable real-time execution of multimodal Physical AI on humanoid robot platforms under strict onboard computational constraints. While recent Vision-Language-Action (VLA) and generative policy models based on transformer or diffusion architectures demonstrate strong manipulation capabilities, their computational complexity and latency limit practical deployment on humanoid robots equipped with embedded GPUs such as NVIDIA Jetson-class systems, especially when rich multimodal sensing (vision, proprioception, and force) is required. This project proposes a lightweight and hierarchical Physical AI architecture, based on software–model–system co-optimization, that integrates multimodal perception, learned policy inference, and low-level control into computationally efficient modules. The proposed approach emphasizes optimizing software and model structures to efficiently map Physical AI components onto available embedded computing resources. These modules can be selectively deployed across onboard computers and, when necessary, partially offloaded to edge or server systems through wireless networks. Target manipulation tasks will be defined and demonstrated through humanoid teleoperation to			

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	collect task-specific multimodal datasets. These datasets will be used to adapt, compress, and extend open-source VLA and learned policy models into efficient, task-specialized multimodal Physical AI policies capable of operating at high control frequencies. The proposed technology will enable humanoid robots to execute Physical AI-driven manipulation at approximately 100 Hz, supporting responsive, stable, and contact-rich whole-body motions suitable for real-world environments.
Rationale for Joint R&D Project	Successful realization of onboard multimodal Physical AI for humanoid robots requires complementary expertise spanning teleoperation-based data generation, learning-based policy development, and efficient system-level architecture design. At UIUC, KIMLAB brings strong capabilities in humanoid teleoperation, multimodal data collection, and learned Physical AI policies, while PI Nam Sung Kim's lab contributes expertise in system architecture, model efficiency, and computational optimization. Together, these strengths enable end-to-end development of lightweight and scalable Physical AI architectures suitable for embedded humanoid platforms. In parallel, Korea is making significant national investments in humanoid robotics, and a growing number of Korean robotics SMEs are developing humanoid platforms and related technologies. Notably, companies are emerging with robots that support whole-body kinesthetic sensing and high backdrivability, as well as dexterous hands with sensing, which are well aligned with multimodal Physical AI research. In addition, Korean startups developing edge AI hardware and embedded AI computing platforms are becoming important enablers of real-time AI on resource-constrained systems. Collaboration between UIUC and Korean SMEs will create strong synergy by combining advanced academic research with practical hardware platforms, application scenarios, and industrial system integration experience. Through this joint R&D effort, the team will co-design efficient multimodal Physical AI policies and distributed computing strategies that balance onboard execution and optional wireless offloading, and validate them on a partner's humanoid robot platform in realistic manipulation tasks. This collaboration will significantly accelerate translation of Physical AI from laboratory demonstrations to deployable humanoid robot systems.
Keywords	1. Multimodal Physical AI 2. Humanoid Robot 3. Lightweight Policy Architecture 4. Vision-Language-Action Model 5. Real-Time Manipulation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

Current Technical Capabilities – PI Joohyung Kim's Lab (KIMLAB)

Professor Joohyung Kim is an Associate Professor in ECE and MechSE at UIUC, where he directs the KIMLAB (Kinetic Intelligent Machine Laboratory). His research focuses on humanoid robot design and control, robot learning systems, and safe human–robot interaction. Professor Kim received his B.S.E. and Ph.D. degrees in Electrical Engineering and Computer Science from Seoul National University in 2001 and 2012, respectively. His career spans academia and industry, including roles as a Research Scientist at Disney Research, a postdoctoral fellow at CMU's Robotics Institute for the DARPA Robotics Challenge, and a Research Staff Member at SAIT (Samsung Advanced Institute of Technology) and Samsung Electronics, where he developed bipedal walking controllers for humanoid robots. He is widely recognized as one of the Korean researchers with the longest and most diverse experience in humanoid robotics across leading universities and global technology companies.

KIMLAB has established strong and sustained capabilities in teleoperation-driven multimodal data collection and learning-based manipulation, centered around PAPRLE (Plug-And-Play Robotic Limb Environment), a modular ecosystem that enables flexible placement and intuitive control of robotic limbs across a wide range of robotic embodiments and application scenarios [1]. PAPRLE is designed to decouple leader devices and follower robots through standardized software and hardware interfaces, enabling rapid reconfiguration of teleoperation setups without redesigning the overall control pipeline. PAPRLE supports a wide range of leader devices, including pluggable puppeteers [CHILD][2], gaming-style controllers, and VR-based interfaces, and provides bilateral teleoperation with real-time force feedback. Depending on the kinematic match between leader and follower, PAPRLE supports both joint-space and task-space control, allowing intuitive demonstration of dexterous manipulation as well as whole-arm motions.

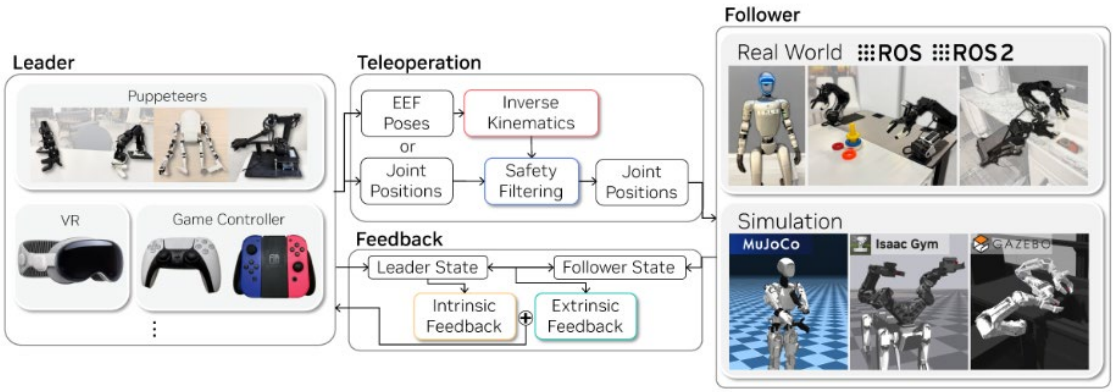


Figure 1 Overview of the teleoperation-based data collection pipeline from KIMLAB's PAPRLE framework.

On the follower side, PAPRLE is built around PAPRAS (Plug-And-Play Robotic Arm System), a portable and modular robotic arm platform developed to enable rapid deployment and experimentation across diverse physical environments. PAPRAS features a lightweight arm with moderate payload capability and modular docking mounts, allowing the same arm to be installed on tables, mobile bases, or custom fixtures [3, 4]. The PAPRAS software stack is fully compatible with PAPRLE, enabling immediate teleoperation, data collection, and policy execution once the arm is connected. This design has allowed KIMLAB to repeatedly validate

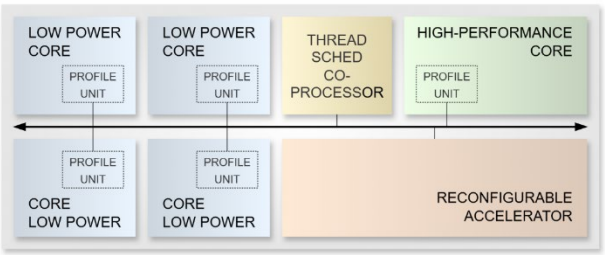
teleoperation and manipulation pipelines across different embodiments without extensive system re-engineering.

Leveraging the plug-and-play leader–follower abstraction, KIMLAB has used PAPRLE to collect scalable teleoperation datasets across multiple embodiments by mixing and matching leader devices and follower robots, including PAPRAS and multiple commercial robot platforms. The collected datasets contain synchronized multimodal signals, including visual observations (RGB and RGB-D), robot proprioception (joint positions, velocities, and torques), force/torque sensing, and other modalities when available. This multimodal data foundation enables learning of contact-rich manipulation behaviors and supports systematic comparison of policy performance across different robot morphologies.

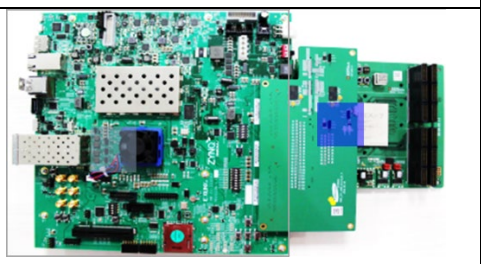
Building on these data capabilities, KIMLAB has actively explored learned manipulation policies using modern imitation and generative policy learning approaches, including Action Chunking Transformer (ACT) and diffusion-based policies. These learned policies have been trained using teleoperated demonstrations and evaluated in both simulation and real-world robotic platforms. Furthermore, KIMLAB has integrated learned policies with the PAPRLE teleoperation framework to support hybrid human-in-the-loop operation, where human demonstrations can initialize policies, policies can autonomously execute actions, and human operators can intervene or refine behaviors when needed. This closed-loop workflow provides an effective mechanism for rapidly iterating between data collection, policy learning, and deployment.

Collectively, these existing technologies, modular teleoperation architectures (PAPRLE), portable robotic arm embodiments (PAPRAS), scalable multimodal dataset collection, and prior experience with learned manipulation policies, provide a strong and practical foundation for the proposed R&D project. They position KIMLAB to extend from task-level learned policies toward lightweight, task-specialized Vision-Language-Action (VLA) based Physical AI architectures that can be executed in real time on humanoid robot and/or other robot platforms under constrained computational resources.

<u>Current Technical Capabilities – PI Nam Sung Kim's Lab (FAST Lab)</u>



Energy-efficient heterogeneous domain-specific



Hardware prototyping capability.

Professor Kim stands as the most prominent figure in computer architecture among Korean researchers in the field. He holds the distinct honor of being the first Korean scholar elevated to the prestigious status of both IEEE Fellow and ACM Fellow in his 40s. Furthermore, he is the first Korean researcher who became a member of the Hall of Fame (HoF) for all three premier computer architecture symposia: ISCA, MICRO, and HPCA. Beyond academia, his distinguished career includes a tenure as Senior Vice President at **Samsung's Memory Business Unit**, where he spearheaded the development of the world's first **HBM-based Processing-In-Memory (PIM)** architecture—a breakthrough that substantially enhances the performance and energy efficiency of emerging AI workloads. This seminal design currently serves as the cornerstone for commercial PIM products slated for market release in late 2026 or early 2027. Additionally, Professor Kim has catalyzed industry advancement by transferring many technologies to leading semiconductor titans through strategic patent licensing and open-source initiatives.

The FAST Lab has pioneered a suite of cutting-edge computing technologies, encompassing the

optimization of state-of-the-art commercial systems for efficient execution of AI workloads, but also the design of general-purpose and domain-specific processors, near-memory processing units, and advanced system software. Transcending the boundaries of theoretical modeling and simulation typically done at academic research labs, the FAST Lab demonstrates its innovations through prototyping across ASIC, FPGA, and integration with commercial platforms.

AI system optimization: The FAST Lab epitomizes excellence in AI systems optimization, pioneering synergistic hardware-software co-designs that address critical bottlenecks in modern machine learning. Our research encompasses the development of high-throughput acceleration for billion-scale retrieval systems through advanced matrix instructions and the optimization of distributed training via in-network lossless compression [CABANA][5]. We possess a unique capability to actualize efficient LLM inference by harnessing on-chip accelerators for rapid decompression, orchestrating cooperative CPU-GPU computation through CXL offloading, and engineering TCO-efficient processing-near-memory platforms [LIA, CXL-PNM][6, 7]. By bridging the gap between theoretical modeling and physical prototyping—including ASIC and FPGA implementations—the FAST Lab delivers transformative solutions that redefine the performance and energy efficiency of emerging AI workloads.

Processor design: The FAST Lab has designed an out-of-order RISC-V processor architecture integrating a reconfigurable spatial accelerator [MESA][8]. This architecture automatically maps code binary based on RISC-V Instruction Set Architecture (ISA) to the reconfigurable spatial accelerator at runtime, accelerating key kernels of AI workloads without rewriting the existing CPU-based code for an accelerator-based code. Later, we extended the design with Samsung to implement a system-on-chip (SoC) using its 7nm technology. We also have an extensive computer architecture simulation capability using gem5, the most widely-used computer architecture simulator. This allows us to evaluate and optimize the performance and energy efficiency of any systems.

Overall Project Objectives

The overall objective of this project is to develop and validate a lightweight, hierarchical, and hardware-aware multimodal Physical AI architecture, based on software–model–system co-optimization, that enables real-time execution of learned perception-to-action policies within the computational constraints of humanoid robot platforms.

Building upon PAPRLE-based teleoperation and multimodal data collection, as well as existing learned policy and Vision-Language-Action (VLA) frameworks, this project will pursue the following specific objectives:

- **Develop lightweight multimodal Physical AI architectures** that significantly reduce inference latency while maintaining robust task performance.
- **Design hierarchical and distributed computing frameworks** that partition perception, high-level reasoning, and low-level control across onboard and optional wireless server resources.
- **Extend teleoperation-driven multimodal data collection pipelines** to support systematic acquisition of large-scale datasets on humanoid platforms.
- **Train and optimize task-specialized multimodal Physical AI policies** using teleoperation datasets.
- **Demonstrate real-time humanoid manipulation** using the proposed architecture on a partner's humanoid robot platform in representative industrial tasks.

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R&D Target Specifications	The project targets the following quantitative and qualitative specifications: Real-Time Performance • End-to-end low-level action generation frequency: ≥ 100 Hz • High-level perception and reasoning update rate: ≥ 10 Hz • End-to-end system latency (sensor input to action output): ≤ 10 ms Embedded Execution Constraints • Onboard GPU memory usage for Physical AI inference: ≤ 16 GB • Onboard power consumption for AI inference: ≤ 30 W • Support for execution on embedded GPU platforms (e.g., Jetson-class systems) Multimodal Capability • Support for multimodal inputs including: vision, proprioception, force/torque, and tactile sensing • Synchronized multimodal data streams with timestamp alignment error: ≤ 2 ms Model Efficiency • Reduction in model size compared to baseline open-source VLA models: ≥ 5× • Reduction in inference latency compared to baseline VLA models: ≥ 5× Task-Level Performance • Task success rate on representative manipulation tasks: ≥ 90% • Generalization across variations in object pose and environment: ≥ 80% success Distributed Execution (when offloading is used) • Wireless communication latency: ≤ 20 ms • Stable operation under intermittent network connectivity Representative tasks include pick-and-place, tool usage, object insertion, and bimanual manipulation under moderate environment variations.

R&D Plan	**<u>Year 1</u>: Teleoperation-Based Multimodal Data Pipeline and Baseline Physical AI Integration**
	• Adapt and extend KIMLAB's existing teleoperation framework to enable intuitive whole-body control of the partner's humanoid robot or manipulation robot.
	• Develop a unified teleoperation pipeline that supports synchronized collection of multimodal data, including:
	  ○ Vision (RGB-D, stereo, or monocular)
	  ○ Robot proprioception (joint positions, velocities, torques)
	  ○ Force/torque sensing
	  ○ Tactile sensing (if available)
	• Integrate the teleoperation system with the partner's robot control and perception stack.
	• Define initial target manipulation tasks and evaluation protocols.
	• Conduct pilot teleoperation experiments to validate data quality, synchronization, and system stability.
	• Benchmark representative open-source VLA and learned policy models on collected pilot datasets and target hardware.
	• Develop baseline task-specialized multimodal Physical AI policies using collected data.
	
	Milestones (Year 1):
	• Teleoperation and multimodal data collection pipeline established.
	• Successful teleoperation of partner's robot for target manipulation tasks.
	• Pilot multimodal dataset collected and validated.

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	• Baseline Physical AI policies demonstrated in laboratory environment.
	
	<u>Year 2</u>: Large-Scale Multimodal Data Collection and Lightweight Physical AI Architecture Development
	• Perform large-scale teleoperation data collection for target tasks using the established pipeline.
	• Curate, organize, and annotate multimodal datasets.
	• Develop lightweight multimodal Physical AI architectures using model compression, distillation, and modularization techniques.
	• Design hierarchical policy structures separating:
	  ○ Multimodal perception and high-level reasoning
	  ○ Low-level action generation and control
	• Implement distributed execution across onboard computers and optional wireless server resources.
	• Evaluate real-time performance, robustness, and task success on hardware.
	
	Milestones (Year 2):
	• Large-scale multimodal dataset constructed.
	• Lightweight Physical AI policies achieving ≥50 Hz low-level control on hardware.
	• Successful execution of multiple target tasks on partner's robot.
	
	<u>Year 3</u>: System Optimization and Pilot-Scale Validation
	• Further optimize model efficiency, memory footprint, and power consumption.
	• Improve robustness under environment variations and contact-rich interactions.
	• Integrate optimized Physical AI system with the partner's humanoid robot platform.
	• Expand task coverage and generalization across object types and scenarios.
	• Conduct pilot-scale validation in SME-provided application scenarios.
	
	Milestones (Year 3):
	• End-to-end control frequency ≥100 Hz achieved.
	• ≥90% task success rate across representative manipulation tasks.
	• Prototype system validated at pilot scale (TRL 6).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The UIUC research team will be responsible for the core algorithmic, system, and architectural development of the proposed multimodal Physical AI technology, including the following roles and responsibilities: • Design and implementation of lightweight and scalable multimodal Physical AI architectures. • Development of hierarchical and distributed policy frameworks that support partitioning of multimodal perception, high-level reasoning, and low-level control across onboard computers and optional wireless server resources. • Development and extension of humanoid teleoperation systems (e.g., PAPRLE) for multimodal data collection. • Design and execution of teleoperation-based experimental protocols for target manipulation tasks. • Collection, curation, synchronization, and annotation of task-specific multimodal datasets. • Training, compression, distillation, and optimization of task-specialized multimodal Physical AI policies. • Integration of trained policies with the partner's humanoid robot platform and control stack. • Performance evaluation, benchmarking, and analysis of real-time execution, robustness, and task success. UIUC will provide software prototypes, trained models, dataset specifications, and technical documentation necessary for technology transfer and joint system integration.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to contribute application-driven development, hardware integration, and pilot-scale validation of the proposed multimodal Physical AI technology, including the following roles and responsibilities: • Provide access to the partner's humanoid robot platform or manipulation robot and detailed hardware and software specifications. ○ Partners with robots that support whole-body kinesthetic sensing and high backdrivability, and/or dexterous hands with kinesthetic sensing and more than 15 degrees of freedom, are preferred. • Define target application scenarios, operational requirements, and success criteria for representative industrial manipulation tasks. • Support integration of the proposed multimodal Physical AI software with the robot's control, perception, and communication systems. • Provide test environments and operational conditions representative of real industrial or service settings. • Participate in joint system debugging, testing, and validation of teleoperation, data collection, and policy execution pipelines. • Support pilot-scale demonstrations and performance evaluation on the partner's robot platform. In addition to robot platform companies, collaboration with Korean SMEs developing edge AI hardware or embedded AI computing platforms (e.g., Rebellions, Furiosa, and DeepX) is encouraged. Such partners are expected to provide hardware platforms, toolchains, and optimization support for deploying the proposed Physical AI software on embedded systems. The SME partner will also contribute engineering resources for hardware interfacing, firmware and driver modification, system optimization, and long-term maintenance, and will actively participate in joint planning, technical reviews, and co-development activities throughout the project.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to produce a deployable multimodal Physical AI technology that enables real-time humanoid manipulation under realistic onboard computational constraints. The primary expected outcomes and impacts include: **Technological Outcomes** • A lightweight, hierarchical, and scalable multimodal Physical AI architecture capable of executing learned perception-to-action policies on humanoid robot platforms with end-to-end control frequencies of approximately 100 Hz. • Task-specialized multimodal Physical AI policies optimized for embedded GPU platforms. • A teleoperation-driven multimodal dataset and standardized training pipeline for humanoid manipulation. • A validated prototype system integrated with a partner's humanoid robot platform and demonstrated at TRL 6. **Commercialization Potential** • The proposed technology can be integrated into commercial humanoid robots as an onboard Physical AI software module. • The technology enables Korean SMEs to develop intelligent humanoid products without reliance on external computing servers or cloud-based inference. • The outcomes can serve as a foundation for licensing, joint product development, and follow-on commercialization and scale-up projects. **Broader Industrial and Economic Impact** • Strengthening the competitiveness of Korean humanoid robot and advanced manufacturing industries. • Reduction in development time and cost for intelligent manipulation systems through reusable data pipelines and modular Physical AI architectures. • Advancement of multimodal Physical AI as a core enabling technology for next-generation robotics.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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